Introduction to Python Programming
MCQ –Unit 1

1. Which of the following is the correct extension of the Python file?
   a) .python
   b) .pl
   c) .py
   d) .p

Answer: c) .py

2. What will be the value of the following Python expression 4 + 3 % 5
   a) 7
   b) 2
   c) 4
   d) 1

Answer: a) 7

3. What is the order of precedence in python?
   a) Exponential, Parentheses, Multiplication, Division, Addition, Subtraction
   b) Exponential, Parentheses, Division, Multiplication, Addition, Subtraction
   c) Parentheses, Exponential, Multiplication, Addition, Division, Subtraction
   d) Parentheses, Exponential, Multiplication, Division, Addition, Subtraction

Answer: d) Parentheses, Exponential, Multiplication, Division, Addition, Subtraction

4. What are the values of the following Python expressions?
   2**(3**2)
   (2**3)**2
   2**3**2

   a) 512, 64, 512
   b) 512, 512, 512
   c) 64, 512, 64
   d) 64, 64, 64

Answer: a) 512, 64, 512

5. Which of the following functions is a built-in function in python?
   a) factorial()
   b) print()
   c) seed()
   d) sqrt()

Answer: b) print()

6. Which of the following is false about Identifiers
    a) Identifiers can be combination of uppercase and lowercase letters, digits or an underscore (_).
    b) An Identifier can start with digit.
    c) We can't use special symbols like !,#,@,%,$ etc in Identifier.
    d) Identifier can be of any length.

Answer: b) An Identifier can start with digit

7. Which of the following is Identity operator
    a) in
    b) not in
    c) not
    d) is

Answer: d) is

8. What does 3 ^ 4 evaluate to?
    a) 81
    b) 12
    c) 0.75
    d) 7

Answer: d) 7

9. Which is the correct operator for power(x^y)?
    a) x^y
    b) x**y
    c) x^^y
    d) x*y

Answer: b) xy**

10. Which of the following operators has its associativity from right to left?
    a) +
    b) //
    c) %
    d) **

Answer: d) **

11. What will be the output of X&Y if x=15 and y=12
    a) b1101
    b) 0b1101
    c) 12
    d) 1101

Answer: c) 12

12. Which of the following represents the bitwise XOR operator?
    a) &
    b) ^
    c) |
    d) !

Answer: b) ^

13. Which of the following is invalid?
    a) _a = 1
    b) _ a = 1
    c) __str _ = 1
    d) str# = 1

Answer: d) str# = 1

14. What is a variable in Python?
    a) A reserved word
    b) A data type
    c) A location in memory to store data
    d) A function

Answer: A location in memory to store data

15. Which of the following is not a valid data type in Python?
    a) int
    b) float
    c) char
    d) str

Answer: c) char

16. Which symbol is used to start a single-line comment in Python?
    a) //
    b) /* */
    c) `
    d) #

Answer: d) #

17. How do you write a multi-line comment in Python?
    a) Using /* and */
    b) Using // at the beginning of each line
    c) Using triple quotes ("""...""" or '''...''')
    d) Using # at the beginning of each line

Answer: c) Using triple quotes ("""...""" or '''...''')

18. How many elif statements can an if-else block contain?
    a) Only 1
    b) Only 2
    c) As many as needed
    d) None

Answer: c) As many as needed

19. Which of these is NOT a valid for loop syntax in Python?
    a) for i in range(5):
    b) for i in [1, 2, 3]:
    c) for i = 0; i < 5; i++:
    d) for i in 'hello':

Answer: c) for i = 0; i < 5; i++:

20. What does range(5) produce?
    a) 1, 2, 3, 4, 5
    b) 0, 1, 2, 3, 4
    c) 5, 4, 3, 2, 1
    d) 0, 1, 2, 3, 4, 5

Answer: b) 0, 1, 2, 3, 4

21. What is the syntax for an if statement in Python?
   a) if condition:
   b) if condition;
   c) if (condition)
   d) if condition then

Answer: a) if condition:

22. What does the elif statement do in Python?
   a) Executes code if the previous if condition is True
   b) Executes code if the previous if condition is False
   c) Executes code if the previous if and elif conditions are False
   d) Creates a new loop

Answer: b) Executes code if the previous if condition is False

23. What does the range() function do in Python?
   a) Creates a list of characters
   b) Creates a list of numbers
   c) Creates a list of strings
   d) Creates a list of dictionaries

Answer: b) Creates a list of numbers

24. What will be the output of the following code?

```
for i in range(5):
    if i == 3:
        break
    print(i)
```

   a) 0 1 2 3 4
   b) 0 1 2 3
   c) 0 1 2
   d) 0 1 2 3 4 5

Answer: c) 0 1 2

25. What will this code output?

```
x = 0
if x:
    print("True")
else:
    print("False")
```

a) True
b) False
c) Error
d) No output

Answer: b) False

26. What is the purpose of a loop in Python?
a) To execute a code block once
b) To stop a program
c) To execute a code block multiple times
d) To check the validity of an expression

Answer: c) To execute a code block multiple times

27. Which keyword is used to exit a loop in Python?
a) exit
b) stop
c) break
d) return

Answer: c) break

28. What does the 'continue' keyword do in a loop?
a) Pauses the loop
b) Stops the loop
c) Skips the rest of the code inside the loop for the current iteration
d) Exits the program

Answer:c) Skips the rest of the code inside the loop for the current iteration

29. Which loop is typically used when the number of iterations is known?
a) For loop
b) While loop
c) Do-while loop
d) Infinite loo

Answer: a) For loop

30. What will be the output of the following code?

```
for i in range(1, 6):
    if i == 3:
        continue
    print(i)
```

a. 1 2
b. 1 2 3
c. 1 2 4 5
d. 1 2 4

Answer: c) 1 2 4 5

31. What will be the output of the following Python code?

```
i = 1
while False:
    if i%2 == 0:
        break
    print(i, end= " ")
    i += 2
```

a) 1
b) 1 3 5 7 …
c) 1 2 3 4 …
d) No Output

Answer: d) **No Output**

32. What will be the output of the following Python code?

```
count = 0
while count < 3:
   print("Hello")
   count += 1
else:
   print("Else block")
```

a. Prints "Hello" three times and then prints "Else block."
b. Prints "Hello" four times.
c. Prints "Else block" three times.
d. Raises a SyntaxError

Answer: Prints "Hello" three times and then prints "Else block."

33. What does the pass statement do in Python?
    a. Exits a loop
    b. Acts as a placeholder that does nothing
    c. Skips the rest of the code in a block
    d. Raises an exception

Answer: b. Acts as a placeholder that does nothing

34. What will be the output of the following code snippet?

```
for i in range(3):
    if i == 1:
        pass
    print(i)
```

    a. 0 1 2
    b. 0 2
    c. 0 0 0
    d. Error

Answer: a. 0 1 2

35. What will be the output of the following Python code

```
num = 10
while num > 0:
   if num % 2 == 0:
      print(num, end=" ")
   num -= 1
```

    a. Prints even numbers in reverse order from 10 to 1.
    b. Prints odd numbers in the range [1, 10].
    c. Prints even numbers in the range [1, 10].
    d. Raises a SyntaxError.

    Answer: Prints even numbers in reverse order from 10 to 1